How we went about this

The modular-degree note, its companion book and its computational record, 4 to 25 October 2026

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consolidated from three session reports and checked against the repositories

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Abstract

In three weeks one author and several AI sessions went from a general brief (search Cremona's tables of elliptic curves for relations that are not already known) to a 39-page research note, *Local factors of the adjoint L-function and the ℓ-adic valuations of the modular degree*, with a main theorem whose inputs are all proved or imported with exact citation, a 1,191-page companion book that prepares a reader for it and lets a specialist referee it, and a computational record that has been rerun in full from pinned data. This account consolidates three draft reports written separately by the sessions involved, which did not know of each other, and checks each statement against the two repositories, their commit histories, the result files and the logs. Where a draft said more than the record supports, this account says what the record shows; Appendix A lists those points.

Contents

1 The parties and the record **2**

2 Timeline **2**

3 The research (4 to 15 October) **3**

4 Planning the book (11 to 15 October) **4**

5 Writing the book (15 to 23 October) **4**

6 Final versions and checks (24 and 25 October) **4**

7 How claims were checked **5**

8 What went wrong, and how it was caught **5**

9 Effort **6**

10 Lessons **6**

11 State and open items **7**

A Corrections to the three draft reports **7**

1 The parties and the record

The author, Dharmaj Soni, set every goal, approved each plan before it ran, chose between options, brought in outside opinions and took the decisions that could not be undone. The work itself (computing, reading, writing, checking) was done by AI sessions (Claude, Anthropic), each of which delegated parts of its work to sub-agents with written briefs. No human expert has yet read the book or the new proofs line by line; the note and the book both say so.

Party	Repository	Dates	Work (commits)
Research session	`mcptrial`, `ecmine/`	4–15 Oct	data, search, conjectures, the note to version 1 (26)
The author	`modular_math_book`	11 Oct	added version 1 of the note (547 lines) (1)
Planning session	`modular_math_book`	11–15 Oct	plan, outlines of all chapters, findings, memo, version 2 of the note (45)
Pilot session	`modular_math_book`	15 Oct	Chapter 10 as a pilot, cost measurement, hand-over (7)
Console run	`modular_math_book`	15 Oct	Chapter 1 (4, without a session tag)
Book session	`modular_math_book`	16–25 Oct	Chapters 2–22, final note, pipeline, full rerun (165)

Sessions are identified by the `Claude-Session` line of each commit. The repositories are https://github.com/dsr001001/mcptrial (branch `ccr-f42a94ab-alw7cu`) and https://github.com/dsr001001/modular_math_book (branch `claude/textbook-paper-prerequisites-clhh5a`, 224 commits).

2 Timeline

Date	Party (commit)	What happened
4 Oct	Research (`2d671af`–`d5c399e`)	Frozen data snapshot and feature table; exclusion list; pre-registered null design; 112 pre-registered tests with 22 controls; the candidate U1; three conjectures checked once on the hold-out.
6 Oct	Research (`ec68583`, `4b64dc7`, `6e3cc59`)	Extension to conductor 500,000; generator of out-of-table curves; first draft of the note; the twist-symmetrised bound replaces the "all but the largest" rule; final set of 893 out-of-table curves.
10 Oct	Research (`3f26e31`–`1f4e045`)	Four outside documents assessed; proof gap in the qualitative theorem closed; local-factor memorandum and the Euler-factor rule with PARI's factors at every bad prime; note restructured; AI statement; by evening the rule is a theorem under the hypothesis of Diamond, Flach and Guo (24 pages, version 1).
11 Oct	Author; Planning	Version 1 added to the book repository. Master plan, LaTeX skeleton, lint and build tools, verified bibliography. A first attempt to outline all chapters at once stopped at a usage limit.
12–14 Oct	Planning	Outlines of all 22 chapters, one to three at a time, each checked and committed; conventions appendix; the published Diamond–Flach–Guo paper read for the central question; memo for the author; 80 findings about the note.
15 Oct	Planning (`ff04c4b`–`5cc47e7`)	Author's decisions D1–D4; version 2 of the note (32 pages); plan PDF; instructions and tools to continue.
15 Oct	Research (`f57c783`)	Review of version 2: one count wrong (28 for 44), the identification of the integral structure in Section 4.1 needs a proof or citation, the AI statement needs rewording.
15 Oct	Pilot; Console	Chapter 10 written and reviewed in three rounds as a pilot, its cost measured; hand-over; Chapter 1 written.

Date	Party (commit)	What happened
16–23 Oct	Book	Chapters 2–22 and both appendices, Chapter 21 (the reading of the note) last; findings grow to 112; 38 points for correction listed in Chapter 21 and a memo (F1–F13); a pass adding explanations; book complete at 1,187 pages.
24 Oct	Book	Final versions in `final/`; three outside reviews assessed; Proposition 4.1 proved; tables printed; pipeline copied and made usable by outsiders; fifteen clarity edits; claim ledger; reviewer's route.
24–25 Oct	Book	Full rerun of the pipeline and recomputation of the numbers with no saved output; two counts corrected; protocol wording made neutral; this report.

3 The research (4 to 15 October)

Plan before computing. The brief was symbolic regression across invariant tables of elliptic curves over $\mathbb{Q}$. A ten-step plan was agreed before anything ran: a frozen snapshot (Cremona's `ecdata` at commit `25cec5e`: 3,064,705 curves in 2,164,260 classes, conductor below 500,000); a feature table with sanity checks; an exclusion list of known relations, each verified on the table; a pre-registered null design with fixed splits (working set $N \leq 300{,}000$, hold-out $300{,}000 < N \leq 400{,}000$ untouched until a single test); a budget of 112 tests with 22 controls; then interpretation, sharpening, one hold-out test, a literature check and a write-up. One positive control (Watkins' bound) failed because a variance statistic cannot see an inequality; rather than lower the bar, a second test family able to see constraints was added, and the failure was reported.

One candidate, pursued to the end. Of the candidates, one was pursued: the ℓ-adic valuation of the modular degree of the optimal curve against local invariants at the bad primes. Its multiplicative part at $\ell \geq 5$ was a known theorem (Kim–Ota, after Agashe–Ribet–Stein); three further statements went beyond it and held on the working set and the hold-out.

State, test, break, restate. From 4 to 10 October a loop ran about a dozen times: state the sharpest rule consistent with the working set, test it on the other ranges (hold-out; the extension to 500,000; 893 curves generated outside the tables up to conductor 3,994,600, with degrees computed from scratch), ask what mechanism would produce it, and restate. Two breaks were decisive. The "all but the largest weight" rule held on all data and was still an artifact of mixing the prime 2 with odd primes; separating them gave the clean twist rule. And a referee's question about the exact local factors led to computing, with PARI's `lfunsympow`, the Euler factor of the adjoint L-function at every bad prime: the local term is the valuation of the factor that the naive adjoint L-function omits, plus a Tamagawa term at potentially multiplicative primes. Every earlier statement became a corollary, and the equality rate at $\ell = 3$ rose from 39.4% to 57.5%. A principle held throughout: an inequality with no violations is cheap, and the equality rate, with the expected number of violations under independence (about 605,000), is what gives it meaning.

Outside reports and the sources. On 10 October the author brought four documents (two referee reports written by AI systems, a note assessing one of them, and an independent review). The rule was to assess before adopting. Factual claims were checked on the data: every exception of the uncorrected rule has mod 3 image the normaliser of a split Cartan subgroup ($364 + 68 + 43 = 475$, shortfall always 1); the reports' explanation by congruences ($q \equiv 1 \pmod{12}$) was wrong and the description became image-based. Claims about the literature were checked against the sources (Agashe–Ribet–Stein's Table 1; the published Diamond–Flach–Guo, Annales ENS 2004, pages 663–727). The rule became a theorem under Diamond–Flach–Guo's hypothesis by writing

down the chain of their statements, with two new lemmas; each lemma was tested on the data first (the Tamagawa term on 46,222 curves: no violation, equality rising from 45.0% to 60.6%).

4 Planning the book (11 to 15 October)

The planning session read version 1 of the note in full and wrote a master plan: the audience (first courses in analysis, linear algebra, complex analysis and statistics), two reading modes (linear, and reference through a card of core results at the head of each chapter), 22 chapters in six parts with their scope, a dependency graph, a map from each notion of the note to chapters, and strict conventions, with a shared preamble, a lint script and a per-chapter build. A first attempt to run the whole pipeline for all chapters at once stopped at a usage limit (reported at about 2.2 million tokens). The replacement was paced by the author: one to three chapters at a time, each outline checked and committed before the next. Each outline agent read the plan, the note and every earlier outline. A single file recorded every finding about the note, each high-impact claim re-derived before it was recorded; it held 80 points at the hand-over. A conventions appendix fixed sixteen conventions and 48 disagreements between outlines. The author took four binding decisions (state the book's stronger results as theorems; length is a guideline; the character χ_{q^*}; the Manin constant positive), and version 2 of the note (32 pages) incorporated the strengthenings, with one step sketched and the data tables left as placeholders.

5 Writing the book (15 to 23 October)

A pilot session wrote Chapter 10 end to end with three review rounds and measured its cost; a Console run wrote Chapter 1; the book session wrote Chapters 2 to 22 and the appendices, at the author's pace, in the cycle writer, one planned review against a common checklist, one fix pass, then build and lint. Computations used PARI/GP 2.17.2 through `cypari2` on Cremona's tables below 10,000 at the pinned commit, checked by SHA-256 sums; they were made with the note in hand, so they are checks, not blind tests. The author supplied the source papers that could not be fetched, and each was read: Flach's theorem needed a minimal-twist hypothesis, Diamond–Flach–Guo's Proposition 1.4(c) is proved by homothety, not by Ihara's lemma, and several citations were corrected. Chapter 21, the reading of the note, was written last and listed 38 points to correct, none invalidating a theorem. A last pass added explanations where the text was terse, with a script confirming that no statement changed.

6 Final versions and checks (24 and 25 October)

Final versions. At the author's request `final/` received the corrected note and a copy of the book that reads it as a single text, without the history of corrections, which stays in the repository.

Three outside reviews. All were produced by AI systems and each was checked claim by claim. The first offered tables built from "the exact numbers already reported in the prose"; 108 of their 124 numbers were not in the note, so they were rejected. Its argument that the weight w_3 is constant on isogeny classes was right but cited the wrong theorem; it was adopted with the right one (Dokchitser–Dokchitser, Theorem 5.4(1)). The third review found a genuine statement error (Theorem 1.5(c) did not exclude I_0^*) and suggested fifteen clarity edits, all made.

Proposition 4.1. The step left sketched in version 2 was proved integrally, without localisation: with compact support, invariants under the covering group lose nothing, because H_c^0 of the cover vanishes and H^1 of a finite stabiliser with coefficients in $\mathbb{Z}$ is 0. An adversarial agent that had not seen the reasoning confirmed three steps, found a gap in the Hecke step and supplied its fix, and

checked the comparison on six levels. The pairing constant is $\frac{1}{2}$ times an ℓ-adic unit, which is all the argument needs.

Data and pipeline. The six table files existed in the research repository and were printed in the note. The pipeline was copied into `final/pipeline`, leaving the research repository untouched, and made usable by others (a reproduction script that stops on any error or count mismatch, a self-test without data, a checksum manifest, provenance and a README). A claim ledger (`final/claims`, 124 claims) records the status and evidence of each claim; a reviewer's route opens the book.

The full rerun. On a two-core machine with 7 GB, calling PARI through `cypari2`, the pipeline was rerun from the pinned data. Every committed summary and all seven table files were reproduced byte for byte (about 40 minutes for the two stages). The thirteen numbers stated in the note without saved output were recomputed: 21 of 23 figures matched and two were wording errors, corrected (19, not 20, exceptions on one range; 18 curves "optimal or not"). The lemma on local factors was checked on all 5,102,273 (optimal curve, prime) pairs and the mod 3 images on all 2,097,505 optimal non-CM curves; PARI 2.15.4, the version of the original runs, agrees with 2.17.2 on all 38,042 curves below 10,000.

The protocol. The commit history of the research repository showed that the twist inequality was formulated after the hold-out counts of the main inequality were known, and that the out-of-table curves were run through each version of the Euler-factor rule as it was written (none failed, none prompted a change). The author chose neutral wording: the note calls these ranges confirmations, not blind tests.

7 How claims were checked

- *Pre-registration, controls and a hold-out* in the search, with failed controls reported.
- *Counts, not adjectives*: every "always" in the note has a count and a result file.
- *Primary sources*: every imported theorem checked against the original, with statement numbers; every quotation of the note in the book matched verbatim by script.
- *Re-derivation before recording*: high-impact claims re-derived independently before they entered the record of findings or the note.
- *Independent checks for high-stakes steps*: a separate agent, instructed to break the proof, for Proposition 4.1.
- *Proved lemmas tested on the data* before publication, with the equality rate as the test.
- *Evidence levels*: proved, imported, conjectural, empirical, computed; and for computations quoted, saved output, regenerated, or independently recomputed (`final/claims`).
- *Outside reports as hypotheses*: each claim checked before adoption; fabricated material rejected.

8 What went wrong, and how it was caught

- *Rules that were artifacts or wrongly explained*: the "all but the largest" rule (caught by separating $q = 2$); the congruence explanation of the exceptions (caught by tabulating $q \bmod 12$ the same day); a weak first argument for the correction term (replaced by a determinant argument).

- *Errors in the planning record*, each caught by a later agent: the Manin constant on the wrong side of a formula, a wrong zeta factor, a wrong reason about the Hecke algebra, a wrong Kodaira type, counts for the wrong conductor range, one overstated finding.
- *Inconsistent conventions* between independently written outlines; fixed by the conventions appendix and a dedicated pass.
- *Mechanical edits*: one regular expression glued two macros together; another deleted about 180 lines of the note, which was rebuilt by replaying the 57 recorded edits of the agent that had corrected it.
- *Statements stronger than the record*: a clean-rerun claim, a population misread (455,707 pairs), two counts, and the blind-test wording; each found by checking against result files or commit history, and corrected.
- *Usage limits and infrastructure*: the all-at-once first attempt stopped at a usage limit; the checker ran out of memory on the full working set; the repository's ignore rules kept the pipeline logs off GitHub until a fresh clone exposed it.
- *Bibliography from memory*: entries added from memory were flagged and later checked against sources where they were available.

9 Effort

Machine time was small: the whole pipeline reruns in under an hour on two cores, and the recomputations of 24 and 25 October took about two and a half hours more. The planning session reported its sub-agents' usage (about 10 million tokens for the outlines, 2.2 million for the stopped first attempt; outlines of 786 to 1,914 lines, 24,421 in all). No session kept a full accounting of model usage; these figures are as reported, not audited.

10 Lessons

1. Agree a plan, an exclusion list and a pre-registered design before computing, and report controls that fail.
2. Pursue one candidate to the end; let breaks point to cleaner statements; judge a rule by its equality rate.
3. Plan centrally and fix conventions before delegating; let each agent read what earlier agents wrote.
4. Pace large work and commit after each step; the all-at-once attempt failed, the staged one survived interruptions.
5. Treat outside reports, and one's own drafts, as hypotheses: check each claim against data and sources.
6. Keep one source per document; version 2 of the note was written where the research repository could not apply its fixes.
7. Write down what was looked at and when; the protocol question could be settled only from commit history.
8. Give every claim a recorded level of evidence and keep the data, scripts and logs in one place that a stranger can rerun.

11 State and open items

Note	39 pages; all inputs of the main theorem proved or imported with citation; six data tables printed and regenerated.
Book	1,191 pages; 22 chapters, two appendices, preface and reviewer's route.
Record	`final/pipeline` (full rerun, logs, self-test, checksum manifest); `final/claims` (124 claims).
Open	a sweep of references and page numbers; a specialist's reading of Proposition 4.1 and of the chain of Diamond–Flach–Guo results; the exact pairing constant (not needed). See `final/TODO.md`.

A Corrections to the three draft reports

The drafts are kept in the book repository: `docs/how_we_went_about_this_core_paper.pdf` (research session), `docs/how_we_went_about_this_planning_session.pdf` (planning session) and an earlier version of this file (book session, in the commit history). Statements checked and found wrong or too strong:

1. *Research draft, clean-clone reproduction* ("re-downloaded the data and rebuilt everything, and the summaries matched"). The logs show the first run's checker failed (script missing from the clone), the second differed in its columns, and the third matched with the checker of that afternoon, before the tables were regenerated. The full reproduction is the rerun of 24 October.

2. *Research draft, "data it has not seen" and "nothing tuned on the hold-out".* True for the conjectures of 4 October. For the Euler-factor rule, the failures of its first version on the hold-out and the extension were seen before the second version, and the out-of-table curves were run through each version; for the twist inequality, the hold-out counts of the main inequality were known when it was formulated.

3. *Research draft, "the revised note was prepared in another session without the repository's LaTeX source".* Version 2 was written by the planning session from the source of version 1, which the author had added to the book repository on 11 October; it was the research repository that lacked the source of version 2.

4. *Research draft, "fifteen curves"*: they are not named anywhere; a sample of 63 curves across all ranges was recomputed instead, all agreeing.

5. *Planning draft, "46 commits" to the hand-over*: 45 commits to the hand-over of 15 October; the 46th is the report itself.

6. *Planning draft, "no chapter text has been written yet"*: true at the hand-over (01:40 UTC on 15 October); the pilot chapter was begun at 03:16 the same day by another session.

7. *Book draft, "writing the book (16–22 October)"* and its implication that one session wrote all chapters: Chapter 10 (pilot) and Chapter 1 were written on 15 October by other runs; the book session wrote Chapters 2–22 from 16 to 23 October. The repository has 224 commits, not about 210 (the earlier count came from a shallow clone).

8. *All drafts*: token figures are as reported by the sessions; none is audited.